Interview for Data Scientist position at GVC

1 VIP Detection

We have provided a data set of fictitious sports betting customers. There is a business need to understand which customers are more likely to be VIPs so that we target these customers in priority with a new specific offer next month. Please use the data made available in the VIP_Train.csv spreadsheet to train a predictive model to provide your best answer to this business problem.

The training set contains 8 features and one binary target variable named *VIP*. Apply your model on the observations in the VIP_Test.csv spreadsheet and insert your best probability estimates that a customer will become a VIP in the column *VIP*.

The primary evaluation metric will be the *Log Loss*. Moreover, please discuss how you would map the predicted probabilities to class labels and how this mapping will affect the business.

Please explain the modelling approach you have taken and give a brief overview of your code during the interview. The total time for the presentation should be about 20 minutes.

Log Loss definition: $\mathcal{L}(y, p) = -(y\log(\mathrm{p}) + (1 - \mathrm{y})\log(1 - \mathrm{p}))$

Please send back within seven days the following:

- Test set predictions
- Code used
- Presentation